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Distribution of rest tremor in patients with Essential Tremor: Does it lateralize with simple kinetic, postural, or intention tremors?

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ABSTRACT

Background: Essential tremor (ET) is characterized by action tremor, although rest tremor may also occur. Despite this, rest tremor has not been studied extensively. A dearth of studies compare the relationship of rest tremor to other types of tremors in ET.

Methods: Two-hundred-and-thirteen ET cases underwent detailed neurological examination, including evaluations of rest, simple kinetic, postural, and intention tremors. A movement disorders neurologist assessed the prevalence of each type of tremor. Each instance of tremor was classified as either symmetrical (present and equally severe on both sides of the body) or asymmetrical (only present on or more severe on one side of the body). Asymmetrical rest tremor that co-occurred with other types of asymmetrical tremors in a given case was further classified as either concordant (i.e., asymmetry on the same side of the body) or discordant (i.e., asymmetry on opposite sides of the body). Chi-square analyses examined the prevalence of concordant versus discordant pairings.

Results: Sixty-two cases (29.2%) had rest tremor while seated. Fifteen (7.4%) had rest tremor while standing. Among asymmetric tremor pairings, concordant pairings occurred 2–2.7 times more frequently than discordant pairings – for pairings involving rest tremor while seated, 73.2% were concordant and 26.8% discordant ($p < 0.001$); for pairings involving rest tremor while standing, 66.7% were concordant and 33.3% were discordant ($p = 0.25$).

Conclusions: Rest tremor often lateralized with other tremor types. These data add to our knowledge of the phenotypic manifestations of ET.

1. Introduction

Essential tremor (ET) is one of the most prevalent neurological diseases [1–3]. Its primary clinical feature is action tremor [4], which is most commonly simple kinetic but often is also postural [4–6]. A substantial number of ET patients also display intention tremor [7].

Other types of tremors may also occur in ET patients [4]. One that is particularly interesting from a mechanistic vantage point is rest tremor. Rest tremor is a hallmark feature of Parkinson's disease (PD), but it is sometimes observed in ET patients as well, in the absence of other hallmark features of parkinsonism (i.e., bradykinesia, rigidity) [8,9]. Prevalence estimates of rest tremor in ET range from 2% in a population-based setting to 46% in a brain bank sample [8,10,11]. Rest tremor is most commonly observed in ET patients with advanced disease (i.e., severe tremor of long duration) [8,10,12]. Rest tremor may also be an early indicator of a second disease process (e.g., PD). Overall, however, there have been very few studies focusing specifically on rest tremor and its clinical features in ET.

Tremors may be either symmetrical or asymmetrical [13]. Symmetrical tremor is present and equally severe on both sides of the body, whereas asymmetrical tremor is only present on or is more severe on one side of the body. Studies of the distribution and laterality of rest tremor in ET and its relations to other types of tremor: (1) are surprisingly rare, (2) have assessed a limited array of tremor types other than rest tremor (i.e., only simple kinetic and not postural or intention), and (3) have yielded somewhat conflicting results. Two studies of ET patients with both rest and simple kinetic tremors reported that the arm characterized by more severe rest tremor was more often ipsilateral to the arm that displayed more severe kinetic tremor (respective N's with rest tremor = 150 and 9) [8,14]. However, a third study (N = 12 with rest tremor) reported the opposite [10].

We expand on earlier work by examining the prevalence and co-

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occurrence of numerous tremor types (i.e., rest, simple kinetic, postural, and intention) in a large cohort of more than 200 ET cases. We were primarily interested in the laterality of rest tremor in ET. Our analysis documents its relations to the laterality of three other types of tremor, filling a gap in current knowledge. If rest tremor lateralizes to the same side as other more common types of tremor in ET, this may suggest a common underlying cause (i.e., that the rest tremor is feature of ET). By contrast, lateralization to the opposite side might suggest different underlying causes (i.e., raising the possibility that rest tremor is a marker of another disease process). An important caveat, of course, is that these data would need to be considered alongside other clinical and phenotypic data.

2. Methods

2.1. Recruited cases

We analyzed data from ET cases enrolled in the Clinical-Pathological Study of Cognitive Impairment in Essential Tremor (COGNET), a prospective, longitudinal brain bank study that enrolls cases 55 and older. Enrollees also met the following criteria: 1) ET diagnosis; 2) no history of brain surgery for ET; 3) willingness to complete neuropsychological tests and videotaped neurological examinations; and 4) willingness to become a brain donor [15,16]. The University of Texas Southwestern Medical Center, Yale University, and Columbia University Institutional Review Boards approved study procedures, and all cases signed informed consents.

2.2. Evaluation, videotaped neurological examination and diagnostic confirmation

Enrolled cases underwent tremor histories and participated in baseline neurological examinations videotaped in their homes and were enrolled from throughout the United States. Examinations were performed by a trained research worker. Evaluation of tandem gait has been previously described [17], as has a detailed cognitive test battery that included an assessment of multiple cognitive domains including memory [17]. All tremor evaluations and ratings described here were completed by an experienced movement disorders neurologist based on a review of the videotaped examination.

As previous work clearly documents marked differences in the prevalence of rest tremor when assessed in different positions [8,18,19], the videotaped examinations included evaluations of rest tremor under two conditions. Following previous researchers [8,19,20], we observed cases while *seated* with hands on their laps (with arms fully supported by the cases' legs), and then while *standing* with arms relaxed and at rest by their side. Rest tremor while seated may be assessed in different hand positions, including semi-prone, completely prone, and hanging down from the armrest [18]. The completely prone position was the method we used because it has been suggested in situations in which one wants the ensure the highest positive predictive value in differentiating ET from PD [18]. Research workers were carefully trained to make sure that the arm was fully supported while resting in the lap to ensure that any observed tremor was truly a rest tremor [14]. Moreover, if the limb did not seem to be fully at rest, the evaluator did not identify rest tremor as present [10].

The neurologist observed and independently assigned each case to one of the following six categories for standing rest tremor and for seated rest tremor: 0 (tremor absent), 1 (tremor present only in dominant arm), 2 (tremor present only in non-dominant arm), 3 (tremor bilaterally present and equally severe), 4 (tremor bilaterally present with dominant arm most severe), and 5 (tremor bilaterally present with non-dominant arm most severe).

The videotaped examination included an assessment of postural tremor. During the examination, cases were videotaped with arms in the outstretched position and again with arms in the wingbeat position. After viewing a given case in both positons, the neurologist provided a single score for postural tremor (i.e., reflecting both positions) for the dominant arm as well as for the non-dominant arm [21,22]. This rating was assigned using the Washington Heights-Inwood Genetic Study of Essential Tremor (WHIGET) tremor rating scale with endpoints of 0 (absolutely no visible tremor) and 3.0 (large amplitude tremor) [5]. Intention tremor was assessed in each arm during the finger-nose-finger maneuver [8], and scored on a 3-item scale as either 0 (absent), 0.5 (probable), or 1 (definite) as per Deuschl et al. [23].

Five tests of simple kinetic tremor (pouring, drinking, using a spoon, finger-nose-finger maneuver, and drawing spirals) [5] were also performed and rated using the WHIGET tremor rating scale. Scores were averaged across the five tasks separately for each arm.

The videotaped neurological examination also included the motor portion of the Unified Parkinson's Disease Rating Scale (UPDRS) [24,25], excluding rigidity and employing a modified rating of rest tremor, as used in this paper. Dystonia was also assessed via views of the face, neck, trunk, and extremities. Finally, to confirm the diagnosis of ET, the movement disorders neurologist reviewed all data and used the WHIGET criteria [26], which are reliable [27] and valid [28] to assign diagnoses of ET. Diagnoses of PD were based on London Brain Bank criteria [29].

2.3. Final sample

Our sample included data from 238 cases who completed the baseline COGNET visit. We excluded three cases assigned diagnoses of PD and 22 assigned diagnoses of dystonia, yielding an final effective sample of 213 ET cases; 132 (62.0%) were female and mean age = 79.6 ± 9.4 years. This sample consisted of longstanding ET cases (mean age of tremor onset = 40.8 ± 22.4 years; mean disease duration = 38.6 ± 22.4 years).

Finally, we classified our 213 ET cases as meeting or not meeting criteria recently proposed for the designation of 'ET-plus' (i.e., impaired gait, impaired memory, questionable dystonic posturing, mild neurological signs of unknown significance; Table 1) [30,31]. Specifically, 177 cases (83.1%) met one or more of the four criteria proposed for such classification.

2.4. Tremor symmetry and concordance categories

The initial six-category classification that was assigned to seated and standing rest tremor (described above, with categories of 0–5) was further collapsed into a four level taxonomy of tremor symmetry. Specifically, a given type of rest tremor was defined as either 1) asymmetrical-dominant (cases in which tremor was initially classified as either present only in the dominant arm, or bilaterally present/more severe in the dominant arm); 2) asymmetrical non-dominant (cases in which tremor was initially classified as present only in the non-dominant arm, or bilaterally present and more severe in the non-dominant arm); 3) symmetrical (previously classified as bilaterally present and equal); or 4) no tremor.

We used the previously described ratings specific to intention and postural tremor to assign cases to parallel symmetry categories for each tremor type (Table 2). Category assignments for simple kinetic tremor were based on the mean of the ratings a given case received on each of the five simple kinetic tremor tasks.

Based on these data, cases were independently assigned to one of the following categories for each type of tremor displayed: 1) asymmetrical-dominant (cases in which a given tremor was present only in the dominant arm, or in which tremor was bilaterally present and *at least 1.0 points higher in rated severity* in the dominant arm); 2) asymmetrical non-dominant (cases in which tremor was present only in the non-dominant arm, or in which tremor was bilaterally present and *at least 1.0 points higher in rated severity* in the non-dominant arm); and 3) symmetrical (tremor bilaterally present and equal, i.e., a *difference in rated severity of less than 1.0 points*); 4) no tremor.

Table 1
ET-plus criteria and classification.

	Meets criteria	Does not meet criteria	Not classified due to missing data
Impaired tandem gait[a]	129 (60.6)	72 (33.8)	12 (5.6)
Impaired memory[b]	102 (47.9)	111 (52.1)	0 (0.0)
Questionable dystonic posturing[c]	3 (1.4)	209 (98.1)	1 (0.5)
Mild neurological signs[d]	99 (46.5)	109 (51.2)	5 (2.3)
Intention tremor[e]	52 (24.4)	158 (74.2)	3 (1.4)
Rest tremor[f]	69 (65.7)	140 (65.7)	4 (1.9)
ET-Plus[g]	177 (83.1)	31 (14.6)	5 (2.3)
ET-Plus[h] (w/out intention tremor)	173 (81.2)	35 (16.4)	5 (2.3)

N(row percentages) of sample either meeting criterion, not meeting criterion, or not classified on a particular criterion due to missing data. N = 213.

[a] Cases walked 10 steps heel-to-toe in a straight line; the number of steps taken off line served as measure of gait impairment. These were compared to the normative number of mis-steps reported in the literature for healthy controls drawn from different age groups. We classified participants with more missteps than the mean of their age group as exhibiting impaired tandem gait.

[b] Cases completed California Verbal Learning Test II Long Delay Free Recall test, the Wechsler Memory Scale Revised: Logical Memory – Delayed, and the Wechsler Memory Scale IV: Verbal Paired Associates Delayed Recall. We averaged the *z*-scores for each test, derived from normative data based on age and education level, to yield an aggregate standardized memory score. Those with a *z*-score less than 0 were classified as displaying impaired memory.

[c] Cases with a clinical diagnosis of dystonia were excluded from the sample. Those displaying mild abnormal postures met this criterion.

[d] Met by those who displayed rest tremor, intention tremor, or both.

[e] Met by those who displayed intention tremor.

[f] Met by those who displayed rest tremor.

[g] Those meeting one or more of these four criteria (with 'mild neurological signs' defined as *either* the presence of rest or intention tremor) were classified as 'ET-plus'.

[h] Those meeting one or more of these four criteria (with 'mild neurological signs' defined as *only* the presence of rest tremor) were classified as 'ET-plus'.

Table 2
Classification of ET cases into tremor asymmetry categories based on distribution of rest, kinetic, postural, and intention tremor.

Category	Type of Tremor				
	Rest Tremor While Seated N = 212	Rest Tremor While Standing N = 202	Simple Kinetic Tremor N = 213	Postural Tremor N = 212	Intention Tremor N = 210
Asymmetrical, Dominant[a]	25 (11.8)	10 (5.0)	74 (34.7)	58 (27.4)	17 (8.1)
Asymmetrical, non-dominant [b]	22 (10.4)	3 (1.5)	75 (35.2)	53 (25.0)	15 (7.1)
Symmetrical [c]	15 (7.1)	2 (1.0)	64 (30.0)	82 (38.7)	20 (9.5)
No tremor	150 (70.8)	187 (92.6)	0 (0.0)	19 (9.0)	158 (75.2)
Total with tremor	62 (29.2)	15 (7.4)	213 (100.0)	193 (91.1)	52 (24.7)

Values are numbers (percentages). Percentages are column percentages.
Total N = 213; n's vary slightly in each column due to incomplete data in 5% or fewer cases.

[a] Asymmetrical-dominant (cases in which tremor was present only in the dominant arm, as well as cases in which tremor was bilaterally present but more severe in the dominant arm).

[b] Asymmetrical non-dominant (cases in which tremor was present only in the non-dominant arm, as well as cases in which tremor was bilaterally present but more severe in the non-dominant arm).

[c] Symmetrical (bilaterally present and equal).

Based on these data, we further classified all *asymmetrical* pairings observed within an individual of (1) of rest tremor, and (2) either intention, simple kinetic, or postural tremor as *concordant* (i.e., dominant/dominant or non-dominant/non-dominant) or *discordant* (i.e., dominant/non-dominant). For example, a case displaying asymmetrical/dominant standing rest tremor and asymmetrical/dominant postural tremor would be classified as concordant for those tremors; a case displaying asymmetrical/non-dominant seated rest tremor and asymmetrical/dominant intention tremor would be classified as discordant for those tremors (Tables 3 and 4).

2.5. Statistical analyses

Mann-Whitney and chi-square tests compared case groups on clinical and demographic variables. Chi-square analyses assessed whether observed asymmetrical pairings of rest tremor and either simple kinetic, postural, or intention tremors were more often concordant or discordant (Tables 3 and 4).

3. Results

3.1. Tremor prevalence

Sixty-two cases (29.2%) displayed rest tremor while seated, and 15 (7.4%) displayed rest tremor while standing; eight of these cases (3.8%) displayed both. Rest tremor was not detected in the lower limbs and no cases had re-emergent tremor. Cases with rest tremor were older than those without (mean age = 83.9 ± 9.5 vs. 77.3 ± 9.5 years, respectively, $p < 0.001$). Cases with rest tremor also reported a longer duration of tremor than did cases without rest tremor (mean = 43.7 ± 24.9 vs. 35.6 ± 20.6 years, respectively, $p < 0.04$), as well as greater disease severity, as measured by tremor severity scores (mean = 22.5 ± 5.1 vs. 19.3 ± 4.7, respectively, $p < 0.001$). Rest tremor cases were also more likely than non-rest tremor cases to display impaired tandem gait, 69.6% versus 57.1%, X^2 (1) = 5.39 p = 0.02. The presence of memory impairment did not vary as a function of whether rest tremor was present or absent, 53.6% versus 45.0% respectively, X^2 (1) = 1.38, p = 0.24. Cases with rest tremor were more likely to display head or voice tremor on examination, 79.7% and 65.2% respectively, than were cases without rest tremor, 57.9% and 42.1%, respectively, X^2 (1)s = 10.72 and 9.29, p's ≤ 0.001. Finally, these groups did not differ in age of onset, years of education, or sex (all p's > 0.05).

For simple kinetic, postural, and intention tremors, 213 (100.0%), 193 (91.1%), and 52 (24.7%) of our cases were classified as displaying each tremor type, respectively (bottom row, Table 2).

Table 3
Prevalence of concordant and discordant severity pairings of rest tremor while seated with simple kinetic, postural, and intention tremors.

Relationship between rest tremor while seated and other tremor type	Concordant[a]	Discordant[b]	p[c]
Rest tremor while seated with simple kinetic tremor	17 (65.4)	9 (34.6)	0.12
Rest tremor while seated with postural tremor	18 (78.3)	5 (21.7)	**0.007**
Rest tremor while seated with intention tremor	6 (85.7)	1 (14.2)	0.06
Total pairings	41 (73.2)	15 (26.8)	**0.001**

Values are numbers (percentages).
Bolded values are statistically significant ($p < 0.05$).

[a] Concordant refers to a dominant/dominant or non-dominant/non-dominant pairing.

[b] Discordant refers to dominant/non-dominant pairing.

[c] Chi-square tests.

Table 4
Prevalence of concordant and discordant severity pairings of rest tremor while standing with kinetic, postural, and intention tremors.

Relationship between rest tremor while standing and other tremor type	Concordant[a]	Discordant[b]	p[c]
Rest tremor while standing with simple kinetic tremor	4 (57.1)	3 (42.9)	0.71
Rest tremor while standing with postural tremor	4 (80.0)	1 (20.0)	0.18
Total pairings	8 (66.7)	4 (33.3)	0.25

Values are numbers (percentages).
There were insufficient data to complete joint analyses of rest tremor standing and intention tremor.
[a] Concordant refers to a dominant/dominant or non-dominant/non-dominant pairing.
[b] Discordant refers to dominant/non-dominant pairing.
[c] Chi-square tests.

3.2. Tremor symmetry and concordance

Table 2 displays the prevalence of asymmetrical/dominant, asymmetrical/non-dominant, and symmetrical tremor classifications for each type of tremor. We then examined each case exhibiting asymmetrical rest tremor, and identified any pairings we could find with either asymmetrical simple kinetic, postural, or intention tremor. The frequencies of these pairings, as well as the prevalence of concordant versus nonconcordant pairings appears in Tables 3 and 4

Of the total pairings that included rest tremor while seated–collapsing across simple kinetic, postural, and intention tremor– 41 of 56 (73.2%) were concordant and 15 of 56 (26.8%) were discordant, $X^2 = 12.07$, $p < 0.001$. Of the total pairings including rest tremor while standing–collapsing across simple kinetic and postural tremor categories– 8 of 12 (66.7%) were concordant and 4 of 12 (33.3%) were discordant, $X^2 = 1.33$, $p = 0.25$. (There were not sufficient cases displaying both rest tremor while standing and intention tremor to allow parallel analyses of these pairings.)

Chi-square analyses revealed a greater prevalence of concordant (78.3%) versus discordant (21.7%) pairings of rest tremor seated and postural tremor ($\chi 2 = 7.35$, $p = 0.007$) (Table 3), and a marginally higher prevalence of concordant (85.7%) versus discordant (14.2%) pairings of rest tremor seated and intention tremor ($\chi 2 = 3.57$, $p = 0.06$) (Table 3). The prevalence of concordant pairings was not significantly greater than that of discordant pairings for any of the remaining tremor pairs. It is important to note that the numbers is many cells were low, and this limited the ability of our chi-square tests to pick up true differences. One can see the trend observed for every type of tremor pair was in the direction of more frequent concordant pairings, which ranged from 1.8 to 4.0 times more common than discordant pairings (Tables 3 and 4).

4. Discussion

We present the first analysis of the co-occurrence and lateralization of rest tremor with respect to a wide range of other tremor types – simple kinetic, postural, and intention - in an ET cohort. Earlier work in this area, some of which we have published, revealed that rest tremor in ET is often observed in the absence of PD and/or related features, is most prevalent among cases drawn from brain bank samples, and is most often present in older, more severe ET cases of long duration [8,10,14]. Our current findings are entirely consistent with these earlier observations; rest tremor was observed in a third (33.0%) of the current sample who were enrolled as brain donors, no rest tremor case was diagnosed with PD, and our cases with rest tremor were significantly older than, and reported longer disease duration and more severe overall tremor than did our cases without rest tremor.

Our current data further reveal that when asymmetric rest tremor is accompanied by a second asymmetrical tremor, those pairings are more often concordant than discordant. The samples of asymmetrical rest tremor cases observed in two of our earlier studies were not of sufficient size to directly address this issue.[10 14] However, our far larger study of 831 ET cases with rest tremor [8] revealed that the arm with the greater rest tremor was more often the arm with the greater action tremor – in 18/36 (50.0%) cases with asymmetric rest tremor, the arm with the greater rest tremor was the arm with the greater action tremor whereas in 11/36 (30.5%) of cases with asymmetric rest tremor, the arm with the greater rest tremor was contralateral to the arm with the greater action tremor [8] This relationship (i.e., more concordant than discordant pairings) is a finding we have reproduced here in a sample of 213 ET cases. In particular, among current cases who exhibit rest tremor in the setting of simple kinetic, postural, or intention tremors, concordant pairings occurred 2–2.7 times more frequently than did discordant pairings (i.e., 73.2% concordant vs. 26.8% discordant for rest tremor while seated; and 66.7% concordant vs. 33.3% discordant for rest tremor while standing). As noted above, small n values in some cells limited the value of formal statistical testing, with greater value obtained through a careful analysis of relative proportions.

These data could provide some clues about the various mechanisms that might give rise to rest tremor in ET. The overall greater preponderance of concordant pairings provides some support, albeit slight, for the notion that rest tremor is in the majority of these cases likely to be a clinical feature of ET (or what the MDS Consensus paper suggests be termed 'ET-plus'). This would be consistent with data from other studies indicating that most ET cases, and most with rest tremor, do not go on to develop another disease (i.e., PD) [32]. Of course, the only way to actually establish a connection between rest tremor lateralization and risk for PD would be through longitudinal follow-up of these cases over a long time period in order to establish which if any converted to ET with PD.

It is likely that there are multiple causes of rest tremor in an ET cohort, and the available research is consistent with this idea. A growing body of clinical, epidemiological, and electrophysiological evidence suggests that rest tremor is exhibited by a segment of ET cases [11,14, 33]. One study, in particular, demonstrated that ET patients with vs. without rest tremor have structural connectivity differences [34]. The patients with rest tremor showed reduced connectivity in a pathway connecting the globus pallidus, caudate, and supplementary motor area, compared to those without rest tremor [34]. Those authors indicated that these findings suggest that there might be underlying differences in brain structure between those with vs. without rest tremor [34]. Furthermore, rest tremor in ET seems to occur in ET patients with more advanced disease (i.e., tremor of greater severity, tremor of longer duration), generally in isolation of parkinsonian features [10]. However, it is important to note that ET patients with rest tremor are especially likely to develop a combination syndrome of ET-PD [9,35,36]. For example, one study found that ET patients with rest tremor are four times more likely than controls to develop PD [32]. Thus, rest tremor in ET may sometimes serve as a signal of emerging comorbidities.

This study has several limitations. First, one might be concerned that cases' limbs were not fully at rest when evaluated for rest tremor. However, we trained our research workers to ensure that cases' limbs were relaxed and fully at rest during the videotaped examination. Moreover, our movement disorders neurologist was careful to distinguish true rest tremor from breakthrough postural tremor during their evaluations. Second, despite the large sample size of more than 200 ET cases, the relatively low prevalence of several tremor types (e.g., rest tremor while standing, intention tremor) meant that statistical testing did not reveal significant differences even in the setting of several substantial group differences. Third, the identification of rest tremor in patients with longstanding concomitant action tremor is not easy, and there is often disagreement among experienced movement disorder specialists. Thus, the data we present need to be considered within a broader context alongside other clinical and phenotypic data. Fourth,

the goal of these analyses was to assess rest tremor *asymmetry* (i.e., the relative difference in tremor severity between right and left sides) in patients with very mild tremor (MDS-UPDRS criteria ratings of 1 or [rarely] 2) [37] rather than strictly quantify and present rest tremor severity ratings using an ordinal scale. Thus, we did not present data, using the MDS-UPDRS, of the precise prevalence of ratings of 1 or 2. However, this did not prevent us from assessing rest tremor asymmetry. Fifth, enrolled cases, evaluated in their homes, were enrolled from throughout the United States, in many instances far from major urban centers. Hence, dopamine transporter scans were not feasible as a means to rule out PD. Nonetheless, the ET cases with rest tremor had no other signs of parkinsonism, making it unlikely that they had emerging PD. Indeed, isolated rest tremor is a well-documented feature of ET. Nor is the sensitivity of dopamine transporter scanning 100%. Hence, we think it is unlikely that there was significant diagnostic misclassification (i.e., missed cases of PD). Lastly, as our cases were seen in their homes, we were limited to clinical as opposed to physiological measures of tremor. Future studies could benefit from addressing our research questions with additional physiological measures and diagnostic tools [10,33,38].

Our study also has numerous strengths. This is the only study, to our knowledge, to examine the nature of asymmetries in pairings of rest and three other tremors in a large ET cohort. Second, a movement disorders neurologist, rather than a general neurologist, used a valid and reliable clinical rating scale to assess tremor during the videotaped examinations. Third, we videotaped the neurological assessments; videotaping allowed the video rater to re-play footage as many times as needed to clarify phenomenology.

As the only analysis of the co-occurrence of rest tremor with three other tremors in ET, these data may provide additional insights into the relationship of rest tremor with other tremor types in ET. The data are cross-sectional. Longitudinal studies would add additional insights into these relationships over time, including how these relations may be moderated by dynamic external factors (e.g., changes in treatment or medication). Such work could, perhaps, provide some predictive utility to the patterns observed.

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Declaration of competing interest

None.

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